The Category of Containers: Morphisms, Products, and Coproducts

Extracted from a machine-checked development

Abstract

We equip containers and their morphisms with the structure of a category **Cont**. Morphisms consist of a forward map on shapes and a *contravariant* map on positions; with this variance, identity and composition laws hold strictly. We then construct binary coproducts and binary products in **Cont**, exhibiting the universal properties in full (computation rules and uniqueness). The proofs of all stated results have been machine-checked; we record only the mathematical content.

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1 Introduction

Containers—pairs (S, P) consisting of a shape type and a family of position types—classify a class of polynomial endofunctors on types via their extensions. Morphisms of containers are maps on shapes together with contravariant maps on positions; these induce natural transformations of extensions, and conversely (the representation theorem).

The present note packages that morphism data into a category **Cont**, and develops its elementary algebra: binary coproducts and binary products. The subtlety throughout is the variance of the position component. Because positions are pulled back, the composite of two morphisms runs position maps *backwards*; once that convention is fixed, the category laws hold by ordinary function associativity, with no transports.

What this note claims. In ordinary mathematical language: containers and shape-and-position morphisms form a category; that category has binary coproducts (shapes a disjoint union, positions case-wise) and binary products (shapes a cartesian product, positions a *coproduct* of the position sets). The universal properties are the expected bijections

$$\mathrm{Hom}(C \oplus D,\, E) \;\cong\; \mathrm{Hom}(C, E) \times \mathrm{Hom}(D, E), \qquad \mathrm{Hom}(R,\, C \times D) \;\cong\; \mathrm{Hom}(R, C) \times \mathrm{Hom}(R, D).$$

2 Containers and morphisms

We work in a single universe **Type**. Recall:

Definition 2.1 (Container)**.** A **container** C consists of a type $\mathrm{Shape}(C)$ of shapes and, for each shape s, a type $\mathrm{Pos}(C)(s)$ of positions. We write $C = (S, P)$ when convenient.

Definition 2.2 (Container morphism)**.** Let $C = (S_C, P_C)$ and $D = (S_D, P_D)$ be containers. A **morphism** $\varphi : C \Rightarrow D$ consists of

1. a map on shapes $u : S_C \to S_D$, and
2. for each $s \in S_C$, a function $g_s : P_D(u(s)) \to P_C(s)$ (contravariant on positions).

We write $\varphi = (u, g)$, and refer to u as the shape component and g as the position component.

Definition 2.3 (Identity morphism)**.** For a container C, the **identity morphism** $\mathrm{id}_C : C \Rightarrow C$ has shape component the identity function on $\mathrm{Shape}(C)$ and position component, at each shape, the identity function on positions.

Definition 2.4 (Composition)**.** Let $\varphi = (u, g) : C \Rightarrow D$ and $\psi = (v, h) : D \Rightarrow E$ be container morphisms. Their **composite** $\psi \circ \varphi : C \Rightarrow E$ (written in applicative order; diagrammatic order is φ then ψ) has

1. shape component $v \circ u$, and
2. position component, at $s \in \mathrm{Shape}(C)$,

$$g_s \circ h_{u(s)} \;:\; \mathrm{Pos}(E)(v(u(s))) \longrightarrow \mathrm{Pos}(C)(s).$$

Shape maps compose forwards; position maps compose backwards.

Remark 2.5. The backwards composition of position maps is forced by the intended semantics: a labelling of C-positions is reindexed first by φ's pullback and then by ψ's, or equivalently by the composite pullback.

Because the type of the position component depends on the shape component, an extensionality principle for morphisms must account for that dependence.

Lemma 2.6 (Extensionality for morphisms)**.** *Let $\varphi, \psi : C \Rightarrow D$ be container morphisms. If the shape components agree and the position components are (heterogeneously) equal, then $\varphi = \psi$.*

Proof. By lean. □

Lemma 2.7 (Pointwise extensionality)**.** *Let $\varphi, \psi : C \Rightarrow D$ with shape components u and v, and suppose $h : u = v$. If for every shape s and every D-position p of $v(s)$,*

$$g_s(p) \;=\; g'_s((h \cdot s) \cdot p)$$

(after transporting p along the action of h at s), then $\varphi = \psi$.

Proof. By lean. □

3 The category of containers

Proposition 3.1 (Left unit)**.** *For every container morphism $\varphi : C \Rightarrow D$,*

$$\varphi \circ \mathrm{id}_C \;=\; \varphi. \tag{1}$$

Proof. By lean. □

Proposition 3.2 (Right unit)**.** *For every container morphism $\varphi : C \Rightarrow D$,*

$$\mathrm{id}_D \circ \varphi = \varphi. \tag{2}$$

Proof. By lean. □

Proposition 3.3 (Associativity)**.** *For morphisms $\varphi : C \Rightarrow D$, $\psi : D \Rightarrow E$, $\chi : E \Rightarrow F$,*

$$\chi \circ (\psi \circ \varphi) = (\chi \circ \psi) \circ \varphi. \tag{3}$$

Proof. By lean. □

Remark 3.4. All three laws hold strictly: the underlying shape-map operations are ordinary function identity and composition, which are unital and associative on the nose, and the backwards position maps inherit the same equalities with no transport.

Definition 3.5 (The category **Cont**)**.** The category **Cont** has

1. objects: containers;
2. morphisms $C \to D$: container morphisms $C \Rightarrow D$;
3. identities and composition as in Definitions 2.3 and 2.4;
4. unit and associativity laws as in Propositions 3.1, 3.2, and 3.3.

4 Binary coproducts

Definition 4.1 (Coproduct of containers)**.** Let $C = (S, P)$ and $D = (T, Q)$ be containers. Their **coproduct** $C \oplus D$ is the container with

1. shapes $S + T$ (the disjoint union), and
2. positions given case-wise: $\mathrm{Pos}(C \oplus D)(\mathsf{inl}\, s) = P(s)$ and $\mathrm{Pos}(C \oplus D)(\mathsf{inr}\, t) = Q(t)$.

Definition 4.2 (Injections)**.** The **left injection** $\iota_C : C \Rightarrow C \oplus D$ has shape component $\mathsf{inl} : S \to S + T$ and position component the identity on $P(s)$ (positions of an inl-shape *are* the C-positions). The **right injection** $\iota_D : D \Rightarrow C \oplus D$ is defined dually via inr.

Definition 4.3 (Copairing)**.** Given morphisms $\varphi : C \Rightarrow E$ and $\psi : D \Rightarrow E$, their **copairing** $[\varphi, \psi] : C \oplus D \Rightarrow E$ has shape component $\mathsf{elim}(u_\varphi, u_\psi)$ (dispatch on the sum) and position component that of φ on inl-shapes and that of ψ on inr-shapes.

Proposition 4.4 (Left computation)**.** *For morphisms $\varphi : C \Rightarrow E$ and $\psi : D \Rightarrow E$,*

$$[\varphi, \psi] \circ \iota_C = \varphi. \tag{4}$$

Proof. By lean. □

Proposition 4.5 (Right computation)**.** *For morphisms $\varphi : C \Rightarrow E$ and $\psi : D \Rightarrow E$,*

$$[\varphi, \psi] \circ \iota_D = \psi. \tag{5}$$

Proof. By lean. □

Proposition 4.6 (Uniqueness / η-rule)**.** *For every morphism $\chi : C \oplus D \Rightarrow E$,*

$$[\,\chi \circ \iota_C,\ \chi \circ \iota_D\,] \;=\; \chi. \tag{6}$$

Proof. By lean. □

Theorem 4.7 (Universal property of the coproduct)**.** *For every cocone $(\varphi : C \Rightarrow E,\ \psi : D \Rightarrow E)$ there is a unique morphism $\chi : C \oplus D \Rightarrow E$ such that $\chi \circ \iota_C = \varphi$ and $\chi \circ \iota_D = \psi$. Explicitly, $\chi = [\varphi, \psi]$; uniqueness follows from Propositions 4.4, 4.5, and 4.6. Equivalently,*

$$\mathrm{Hom}_{\mathbf{Cont}}(C \oplus D,\ E) \;\cong\; \mathrm{Hom}_{\mathbf{Cont}}(C, E) \times \mathrm{Hom}_{\mathbf{Cont}}(D, E) \tag{7}$$

naturally in E.

Proof. By lean. □

5 Binary products

The product of containers is dual in the categorical sense, but the combinatorics of positions are not a naive product: the positions of a product shape are the *coproduct* of the two position sets. This is the product of the underlying polynomials.

Definition 5.1 (Product of containers)**.** Let $C = (S, P)$ and $D = (T, Q)$ be containers. Their **product** $C \times D$ is the container with

1. shapes $S \times T$, and
2. positions of (s, t) given by $P(s) + Q(t)$.

Definition 5.2 (Projections)**.** The **first projection** $\pi_C : C \times D \Rightarrow C$ has shape component the first projection $S \times T \to S$, and position component (contravariant) the coproduct injection $\mathsf{inl} : P(s) \to P(s) + Q(t)$. The **second projection** π_D is defined dually via inr.

Remark 5.3. The variance is the teaching point dual to the coproduct. There the injections were identity-on-positions; here the projections carry a coproduct injection *backwards* on positions: a C-position is sent into the larger position set $P(s) + Q(t)$.

Definition 5.4 (Pairing)**.** Given morphisms $f : R \Rightarrow C$ and $g : R \Rightarrow D$, their **pairing** $\langle f, g\rangle : R \Rightarrow C \times D$ has shape component $r \mapsto (u_f(r),\ u_g(r))$ and position component, at r, the coproduct elimination of f's and g's position maps.

Proposition 5.5 (Left computation)**.** *For morphisms $f : R \Rightarrow C$ and $g : R \Rightarrow D$,*

$$\pi_C \;\circ\; \langle f, g\rangle \;=\; f. \tag{8}$$

Proof. By lean. □

Proposition 5.6 (Right computation)**.** *For morphisms $f : R \Rightarrow C$ and $g : R \Rightarrow D$,*

$$\pi_D \;\circ\; \langle f, g\rangle \;=\; g. \tag{9}$$

Proof. By lean. □

Proposition 5.7 (Uniqueness / η-rule)**.** *For every morphism $\chi : R \Rightarrow C \times D$,*

$$\langle\, \pi_C \circ \chi,\ \pi_D \circ \chi\,\rangle \;=\; \chi. \tag{10}$$

Proof. By lean. □

Theorem 5.8 (Universal property of the product)**.** *For every cone $(f : R \Rightarrow C,\ g : R \Rightarrow D)$ there is a unique morphism $\chi : R \Rightarrow C \times D$ such that $\pi_C \circ \chi = f$ and $\pi_D \circ \chi = g$. Explicitly, $\chi = \langle f, g \rangle$; uniqueness follows from Propositions 5.5, 5.6, and 5.7. Equivalently,*

$$\mathrm{Hom}_{\mathbf{Cont}}(R,\, C \times D) \;\cong\; \mathrm{Hom}_{\mathbf{Cont}}(R, C) \times \mathrm{Hom}_{\mathbf{Cont}}(R, D) \tag{11}$$

naturally in R.

Proof. By lean. □

6 Discussion

What has been established. Containers and their morphisms form a category **Cont** with strictly holding unit and associativity laws. Binary coproducts and products exist and satisfy the full universal properties. The combinatorics are those of polynomial functors: coproduct of containers is case-wise on both shapes and positions; product of containers multiplies shapes and *adds* positions.

Variance. The entire development turns on the contravariance of the position component. With positions pulled back, composites and universal maps typecheck without transport, and all category and universal-property laws reduce to function extensionality and case analysis on sum types.

Scope. This note does not treat the composition product of containers (the monoidal structure whose comonoids are directed containers), nor the extension functor $\mathbf{Cont} \to [\mathbf{Type}, \mathbf{Type}]$ as a fully faithful embedding. Those belong to companion developments.

Machine checking. Every definition and theorem statement above corresponds to a declaration in a machine-checked formalisation. The phrase "by lean" indicates that the proof has been verified by a proof assistant.

Acknowledgements

The mathematical content follows the theory of containers of Abbott, Altenkirch, and Ghani. The presentation here is a prose rendering of a formal development intended for human review of the theorem statements.

References

[1] M. Abbott, T. Altenkirch, and N. Ghani. Categories of containers. In *FoSSaCS*, volume 2620 of *LNCS*, pages 23–38. Springer, 2003.

[2] M. Abbott, T. Altenkirch, and N. Ghani. Containers: constructing strictly positive types. *Theoretical Computer Science*, 342(1):3–27, 2005.